Semantics-Consistent Representation Learning for Remote Sensing Image–Voice Retrieval

Hailong Ning, Bin Zhao, and Yuan Yuan, *Senior Member, IEEE*

***Abstract*— With the development of earth observation technology, massive amounts of remote sensing (RS) images are acquired. To find useful information from these images, cross-modal RS image–voice retrieval provides a new insight. This article aims to study the task of RS image–voice retrieval so as to search effective information from massive amounts of RS data. Existing methods for RS image–voice retrieval rely primarily on the pairwise relationship to narrow the heterogeneous semantic gap between images and voices. However, apart from the pairwise relationship included in the data sets, the intramodality and nonpaired intermodality relationships should also be considered simultaneously since the semantic consistency among nonpaired representations plays an important role in the RS image–voice retrieval task. Inspired by this, a semantics-consistent representation learning (SCRL) method is proposed for RS image–voice retrieval. The main novelty is that the proposed method takes the pairwise, intramodality, and nonpaired intermodality relationships into account simultaneously, thereby improving the semantic consistency of the learned representations for the RS image–voice retrieval. The proposed SCRL method consists of two main steps: 1) semantics encoding and 2) SCRL. First, an image encoding network is adopted to extract high-level image features with a transfer learning strategy, and a voice encoding network with dilated convolution is devised to obtain high-level voice features. Second, a consistent representation space is conducted by modeling the three kinds of relationships to narrow the heterogeneous semantic gap and learn semantics-consistent representations across two modalities. Extensive experimental results on three challenging RS image–voice data sets, including Sydney, UCM, and RSICD image–voice data sets, show the effectiveness of the proposed method.**

***Index Terms*— Heterogeneous semantic gap, remote sensing (RS) image–voice retrieval, semantics-consistent representation.**

Manuscript received August 24, 2020; revised December 14, 2020, January 22, 2021, and February 16, 2021; accepted February 17, 2021. Date of publication March 5, 2021; date of current version December 9, 2021. This work was supported in part by the National Key Research and Development Program of China under Grant 2020YFB2103902, in part by the National Science Fund for Distinguished Young Scholars under Grant 61825603, in part by the Key Program of National Natural Science Foundation of China under Grant 61632018, and in part by China Postdoctoral Science Foundation under Grant 2020TQ0236. *(Corresponding author: Yuan Yuan.)*

Hailong Ning is with the Shaanxi Key Laboratory of Ocean Optics, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an 710119, China, also with the University of Chinese Academy of Sciences, Beijing 100049, China, and also with the School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, China.

Bin Zhao and Yuan Yuan are with the School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, China (e-mail: y.yuan1.ieee@qq.com).

Digital Object Identifier 10.1109/TGRS.2021.3060705

I. Introduction

With the rapid development of remote sensing (RS) technology, tons of RS images have been produced [1]–[3]. There is no doubt that mining useful information from these RS images [4] is very important. However, considering that the RS data are too large, it is unpractical to find the useful information manually due to the time-consuming workload. Driven by the practical demand, many researchers pay attention to the large-scale RS data retrieval task [2], [4]–[6]. It can automatically refine precise semantic information contained in the RS data and has wide application prospects in military targets detection, military audio intelligence generation, and disaster monitoring scenarios [4], [7], [8].

Generally, the RS data retrieval methods can be roughly divided into two categories: unimodal retrieval and cross-modal retrieval. Specifically, the unimodal RS data retrieval methods conduct the similarity search in the same modality [4], [9]. For instance, Li *et al.* [4] presented a large-scale RS image retrieval method to retrieve the RS images with similar semantics. By comparison, the source and target data are from different modalities in cross-modal RS data retrieval methods [7], [10]. For instance, Chaudhuri *et al.* [7] proposed a deep neural network to learn a discriminative shared feature space for the input RS images and the corresponding voice-based annotations. This article focuses on the cross-modal RS data retrieval, which is more challenging than unimodal RS data retrieval due to the heterogeneous semantic gap of cross-modal data.

As shown in Fig. 1, the task of RS image–voice retrieval is to find the relevant RS image (voice) given a queried voice (image). Notice that the task is more convenient for humans than the way by keyboard inputting text to retrieve similar RS images because it is based on human speeches to retrieve target RS images [10], [11]. As a consequence, the task is more practical in some scenarios that are emergent and not convenient for keyboard input, such as military targets detection, military audio intelligence generation, and disaster monitoring [8], [10].

Due to the practicability of the RS image–voice retrieval task, several related works have been developed [7], [10]–[13]. These works can be classified into two categories: CNN feature-based methods and hash-based methods. The former methods conduct the retrieval task by learning CNN features and their consistency relationship [9]. As an example, Mao *et al.* [10] proposed a deep visual–audio network to learn

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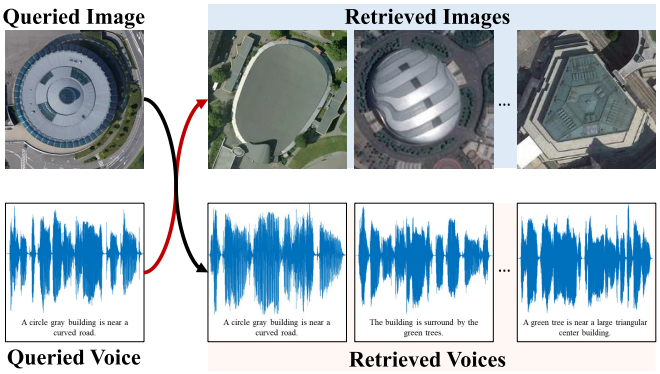


Fig. 1. Schematic of the RS image–voice retrieval task. The goal of the task is to leverage the queried RS image (voice) to retrieve the relevant voices (images).

the correspondence between RS images and voices. The latter methods encode high-dimensional data points into compact binary code for retrieval [14]. For instance, Chen and Lu [12] and Chen *et al.* [13] presented to integrate the feature learning and hash code learning into a unified framework to achieve fast retrieval property between RS images and voices.

Practically, these works commonly narrow the distance between representations from the paired multimodalities directly, resulting in the similar paired representations for cross-modal retrieval. However, the obtained similar representations by these works are difficult to effectively alleviate the pervasive heterogeneous semantic gap because the cross-modal retrieval aims to retrieve all the data with the same semantic concept rather than only one data coming from the same pair. As a consequence, it is necessary to fully mine the relationships of both paired and nonpaired data within and across modalities. Naturally, we propose to simultaneously model the semantic relationships, including pairwise, intramodality, and nonpaired intermodality relationships. Specifically, the pairwise relationship models the consistency between the paired image and the voice. The intramodality relationship is to enhance the consistency between two representations from the same modality and belonging to the same semantic concept, thereby avoiding the adverse impact caused by the problems of interclass similarity and intraclass diversity. The nonpaired intermodality relationship is to enhance the consistency between two nonpaired representations from different modalities but belonging to the same semantic concept. In order to further improve the retrieval performance, the pairwise, intramodality, and nonpaired intermodality relationships should be considered comprehensively.

Inspired by the discussion above, a semantics-consistent representation learning (SCRL) method is proposed to make full use of these three relationships for the RS image–voice retrieval. As shown in Fig. 2, the proposed SCRL method can be divided into two steps: 1) semantics encoding and 2) SCRL. First, the VGG16 network [15] is adopted to automatically extract high-level RS image features with a transfer learning strategy. Meanwhile, the input voices are processed as mel-frequency cepstral coefficient (MFCC) features [16] and input into a 1-D dilated convolutional network to automatically extract high-level voice features. Second, to explicitly compute the similarity between representations from different modalities, a semantics-consistent representation space is explored by considering three kinds of relationships simultaneously. The relationships include pairwise, intramodality, and nonpaired intermodality relationships. They are measured by the consistency loss to narrow the heterogeneous semantic gap across two modalities. In addition, to further learn significantly discriminative and more compact semantic representations, a classification loss is adopted for each branch. Afterward, the consistency loss and the classification loss are combined jointly to learn ultimate semantics-consistent representations for the RS image–voice retrieval.

To sum up, the main contributions can be summarized into threefolds.

1) A novel RS image–voice retrieval method, SCRL, is proposed to fully explore the pairwise, intramodality, and nonpaired intermodality relationships simultaneously. Then, semantics-consistent representations across the two modalities can be learned to effectively alleviate the pervasive heterogeneous semantic gap.
2) An effective voice encoding network is proposed to learn high-level voice features. The network can capture the long-range correlation of voice signal by introducing the 1-D dilated convolutional kernel.
3) Experimental results on three challenging RS image–voice data sets show that the proposed SCRL method can significantly improve the performance of the RS image–voice retrieval. Especially, the mean average precision (mAP) value can be improved by nearly 9% compared with the state of the arts.

The remainder of this article is organized as follows. Section II introduces the related works about the task of RS image–voice retrieval. Section III elaborates the proposed SCRL method. Section IV shows and analyzes the experimental results. Finally, Section V presents the conclusion.

II. Related Works

In this section, the related works about this article are primarily restricted on unimodal and cross-modal RS data retrieval methods.

A. Unimodal RS Data Retrieval Methods

Unimodal RS data retrieval methods aim to search similar RS data in the same modality. Some early works are based on the handcrafted feature [17]–[19]. For instance, Luo *et al.* [17] presented a method by comparing multiple-resolution wavelet features for satellite images retrieval. Yang and Newsam [18] leveraged local-invariant features for RS image retrieval. Aptoula [19] proposed to apply global morphological texture descriptors for RS image retrieval. However, these methods are not suitable for large-scale RS image retrieval since they are based on low-level handcrafted features. With the rapid development of artificial intelligent technology [20]–[24], a large number of unimodal RS data retrieval methods based on deep learning have emerged. For instance, Tang *et al.* [25] developed a two-stage reranking method to improve the

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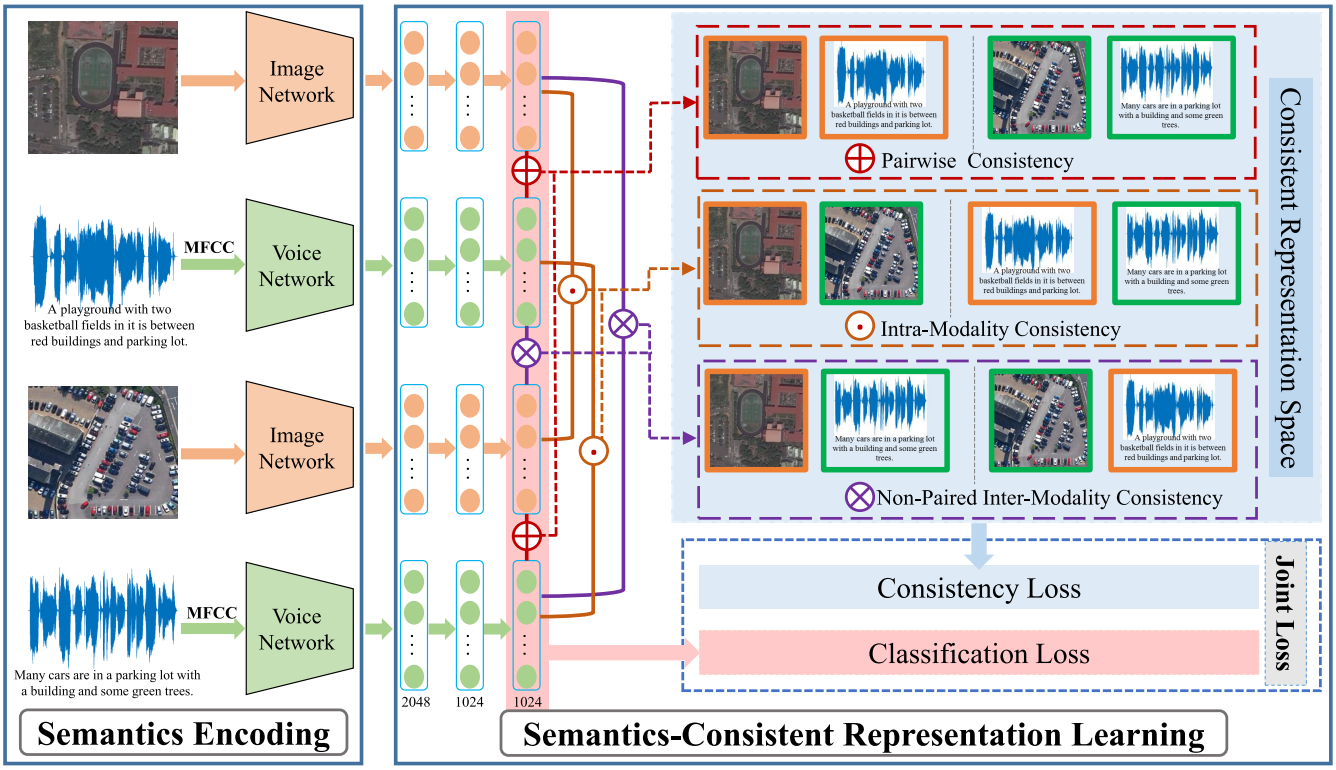


Fig. 2. Overall framework of the proposed SCRL method. First, the image encoding network is adopted to extract high-level image features, and the voice encoding network is devised to obtain high-level voice features. Second, the extracted image features and voice features are fed into the representation step to explore the consistent representation space. By exploring the space, the heterogeneous semantic gap is narrowed and the semantics-consistent representations across two modalities are learned by comprehensively considering the pairwise consistency ($\oplus$), intramodality consistency ($\odot$), and nonpaired intermodality consistency ($\otimes$). MFCC represents the operation to extract the MFCCs feature of the input voice.

retrieval performance. Shao *et al.* [9] designed a fully convolutional network to solve the problem of poor retrieval performance due to multiple labels for single images. Ye *et al.* [26] utilized a weighted distance as similarity criteria to learn determinative representations for the RS image retrieval. To address the RS image retrieval problem due to multiple land-cover classes, Chaudhuri *et al.* [27] introduced a semisupervised graph-theoretic method with only a small number of training images characterized by multilabels. Kang *et al.* [28] proposed a graph relation network to preserve the complex semantic relations pervading RS scenes for multilabel RS image retrieval. In response to the scarcity of labeled images, Liu *et al.* [29] presented an unsupervised deep transfer learning method based on the similarity learning for the RS image retrieval. Ye *et al.* [30] proposed to extract the domain-invariant feature for the RS image retrieval using an unsupervised domain adaptation model. To obtain the quicker response during retrieval, several techniques consider binary features [4], [5], [31]. Song *et al.* [31] developed an image retrieval and compression method based on binary generative adversarial network. Demir and Bruzzone [5] introduced two kernel-based nonlinear hashing methods for the RS image retrieval to reduce the amount of memory required. Li *et al.* [4] introduced an RS image retrieval method, which integrates the feature learning network and hashing network into a unified framework.

B. Cross-Modal RS Data Retrieval Methods

The objective of cross-modal RS data retrieval methods is to search similar RS data across different modalities. Due to the urgent demand for RS data analysis, cross-modal RS data retrieval attracts increasing attention from researchers. Li *et al.* [2] proposed an end-to-end training method to learn source-invariant features for the cross-source RS image retrieval. Xiong *et al.* [32] presented a discriminative distillation network to eliminate the inconsistency across different modalities. To develop an effective retrieval system in harsh environments, Xiong *et al.* [33] designed another deep cross-modality hashing network for the optical and SAR RS images retrieval. Recently, a more natural cross-modal RS data retrieval method has emerged, which is based on RS images and human voices. As the first attempt for the task of RS image–voice retrieval task, Mao *et al.* [10] devised a deep visual–audio network to learn the correspondence between RS images and voices. Subsequently, they develop another cross-modal RS image–voice retrieval method to further improve the retrieval performance [11]. Based on the works of Guo *et al.*, Chen and Lu [12] presented to leverage the triplet similarity of deep features to solve the insufficient utilization problem for the relative semantic similarity relationships. In addition, Chen *et al.* [13] integrated the feature learning and hash code learning into a unified framework to achieve fast retrieval property. As another implementation, Chaudhuri *et al.* [7]

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proposed a deep neural network to learn a common embedding of RS images and spoken words rather than complete spoken sentences. Song *et al.* [34] adopted an intermedia hashing (IMH) model for exploring the data correlations from heterogeneous data sources.

This article is dedicated to conduct the cross-modal RS image–voice retrieval by fully considering the semantic relationships among representations across different modalities. The proposed method comprehensively improves the pairwise, intramodality, and nonpaired intermodality consistency so as to narrow the heterogeneous semantic gap and learn semantics-consistent representations for RS image–voice retrieval.

III. Proposed Method

The proposed SCRL method is composed of two steps: 1) semantics encoding and 2) SCRL. In this section, the overall framework of the proposed method is described first. Second, the semantics encoding and SCRL steps are elaborated. Third, the loss functions are presented. Finally, the optimizing strategy is introduced.

A. Overall Framework

As shown in Fig. 2, the proposed SCRL method consists of four parallel branches and can be divided into two steps, including semantics encoding and SCRL steps. Two of the branches are related to the image modality, and the rest are related to the voice modality. The branches, which are responsible for the same modality, share the same parameters. As a consequence, the features from the same modality own the same transforms. After each iteration, four features from the two modalities are obtained by the semantics encoding step. Then, the obtained features are used as the input of the SCRL step to learn four representations. During this step, the relationships of different representations are explored in a defined consistent representation space. The relationships include pairwise, intramodality, and nonpaired intermodality relationships. All the relationships are sought by calculating the distance between two representations to preserve the semantic consistency across different modalities. Specifically, the pairwise relationship is built to make the semantic information consistent for the paired RS image voice. The intramodality relationship is considered to keep the semantic consistency within each modality. The nonpaired intermodality relationship is developed to enforce the semantic information to be consistent across different modalities for unpaired samples. Afterward, the three kinds of relationships are measured by a consistency loss. Finally, we adopt a joint loss function to simultaneously minimize the consistency loss of the consistent representation space and the classification loss of each branch.

The formal definitions in this article are given as follows. Let $\mathcal{D} = \{(\mathbf{X}_m^I, \mathbf{X}_m^V, \mathbf{y}_m)\}_{m=1}^N$ be the collection of N instances of RS image–voice pairs, where $\mathbf{X}_m^I$ is the mth input image sample, $\mathbf{X}_m^V$ is the mth input voice sample, and $\mathbf{y}_m$ is the mth semantic label. The goal of the semantics encoding step is to learn two mapping functions $\mathcal{F}_I$ and $\mathcal{F}_V$ for extracting high-level image features $\{\mathbf{s}_m^I\}_{m=1}^N$ and voice features $\{\mathbf{s}_m^V\}_{m=1}^N$. Afterward, the SCRL step aims to learn two mapping functions $\mathcal{R}_I$ and $\mathcal{R}_V$ for obtaining the semantics-consistent image representations $\{\varphi_m^I\}_{m=1}^N$ and voice representations $\{\varphi_m^V\}_{m=1}^N$. The obtained semantics-consistent representations are exploited for RS image–voice retrieval.

B. Semantics Encoding

As shown in Fig. 2, the semantics encoding step consists of four parallel networks, in which the two image encoding networks share the same parameters and the two voice encoding networks share the same parameters. The details about image encoding networks and voice encoding networks are described as follows.

1) Image Semantics Encoding: According to the previous works [35], [36], the high-quality feature is quite important for the retrieval task. CNN features have been proved to be high efficiency for the retrieval task because of the success in expressing the high-level semantic information. To learn high-quality CNN features, massive amounts of data with manual annotations, such as ImageNet, are necessary. Unfortunately, large-scale data set with manual annotation in the RS domain is unavailable. In order to solve this problem, a transfer learning strategy has emerged. The transfer learning strategy applies knowledge learned from one domain to other domains [37]. In this article, the transfer learning strategy is leveraged to extract high-quality image features. To be more specific, the VGG16 network [15] pretrained on ImageNet is adopted to acquire the RS image features since some useful texture features may share in natural images and RS images [38].

When using the pretrained VGG16 network for image semantics encoding, the fully connected layers are removed to obtain high-level image features. Following the previous work [39], the network is normalized by scaling weights to ensure the mean activation of each convolutional filter over images and positions to be one. Specifically, the mean activation s_i^l of each convolutional filter in the lth layer over the whole training image set $\mathcal{X}$ and all M_l spatial locations can be calculated as follows:

$$s_i^l = \frac{1}{NM_l} \sum_{\mathcal{X}} \sum_{j=1}^{M_l} \text{relu}\left(w_i^l * x_j^{l-1} + b_i^l\right) \tag{1}$$

where w_i^l represents the weight of ith convolutional filter in the lth layer, x_j^{l-1} stands for the jth patch in the $(l-1)$th layer, $*$ indicates the convolutional operation, b_i^l denotes the bias term in the lth layer, and N is the total number of the training images. Then, the parameters w_i^l and b_i^l are scaled by $(1/s_i^l)$ as follows:

$$\mathbb{E}_{\mathcal{X},j}\left[\text{relu}\left(\frac{w_l^i}{s_i^l} * x_j^{l-1} + \frac{b_l^i}{s_i^l}\right)\right] = 1. \tag{2}$$

According to the above process, the normalization is conducted from bottom layers to top layers sequentially. The process of image semantics encoding can be written as

$$\mathbf{s}_m^I = \mathcal{F}_I\left(\mathbf{X}_m^I; \theta_I\right) \tag{3}$$

where $\mathbf{s}_m^I \in \mathbb{R}^{h_1 \times w_1 \times c_1}$ represents the extracted semantics feature of the mth image $\mathbf{X}_m^I$, $\mathcal{F}_I$ indicates the mapping

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function from image to the corresponding feature, and θ_I is the parameter.

2) Voice Semantics Encoding: Voice belongs to a continuous 1-D signal and the word in it cannot be distinguished easily by computer. In order to quantify the continuous voice, MFCCs feature is first extracted to represent the voice signal according to previous works [16], [40]. Here, the MFCC feature is expressed as $\mathbf{X}_m = \Upsilon(\mathbf{X}_m^V)$.

The obtained MFCC feature is reshaped and then fed into the voice encoding network to perform high-level voice features. To capture long-range correlations, 1-D dilated convolutional kernels are adopted to construct the voice network. Specifically, the voice network is composed of five cascaded 1-D convolutional layers and pooling layers, where each convolutional layer adopts dilated convolutional kernel. The flattened MFCC feature is fed into the voice network and the learned high-level voice feature can be denoted as

$$\mathbf{s}_m^V = \mathcal{F}_V(\mathbf{X}_m; \theta_V) \tag{4}$$

where $\mathbf{s}_m^V \in \mathbb{R}^{h_2 \times w_2 \times c_2}$ represents the extracted high-level feature of the mth MFCC feature $\mathbf{X}_m$, $\mathcal{F}_V$ indicates the mapping function from MFCC feature to the corresponding high-level voice feature, and θ_V is the parameter.

C. Semantics-Consistent Representation Learning

This step aims to learn semantics-consistent representations by comprehensively considering the pairwise, intramodality, and nonpaired intermodality relationships across different modalities. As shown in Fig. 2, these three relationships are measured by a consistency loss to model the similarity of the learned representations. In addition, a classification loss is adopted for each branch to enhance the semantic discrimination ability of representations, resulting in more compact representations. The consistency loss and classification loss are combined jointly to learn semantics-consistent representations for the RS image–voice retrieval. Hereinafter, the details about the SCRL are dwelled on.

1) Architecture of SCRL : As shown in Fig. 2, the SCRL step contains four parallel branches. Similar to the semantics encoding step, two of the branches are related to the image modality, and the rest are related to the voice modality. Actually, the branches responsible for the same modality share the same weights. Each branch is constructed by three cascaded fully connected layers. The specific parameters of each branch are shown in Fig. 2. The features from the semantics encoding step are taken as inputs for the SCRL step. Then, the representations are output from each branch in the representation learning step.

As for the image branch, the learned image representation can be expressed as

$$\varphi_m^I = \mathcal{R}_I\left(\mathbf{s}_m^I; \Theta_I\right) \tag{5}$$

where $\varphi_m^I \in \mathbb{R}^{d_1}$ represents the learned semantics-consistent representation for the mth image, $\mathcal{R}_I$ indicates the mapping function from the image feature to the corresponding semantics-consistent representation, and Θ_I is the parameter.

As for the voice branch, the learned voice representation can be expressed as

$$\varphi_m^V = \mathcal{R}_V\left(\mathbf{s}_m^V; \Theta_V\right) \tag{6}$$

where $\varphi_m^V \in \mathbb{R}^{d_2}$ represents the learned semantics-consistent representation for the mth voice, $\mathcal{R}_V$ indicates the mapping function from the voice feature to the corresponding semantics-consistent representation, and Θ_V is the parameter.

2) Pairwise Consistency: The pairwise consistency represents the semantic information of representations from the paired RS image–voice should be consistent since they describe the same semantic concept in the applied RS image–voice data sets. To measure the similarity of representations, the cosine distance is adopted as it is commonly used in cross-modal retrieval [41], [42]. The cosine distance can be defined as

$$D(\mathbf{x}, \mathbf{y}) = \mathbf{1} - \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|} \tag{7}$$

where $\mathbf{x}$ and $\mathbf{y}$ represent two vectors with the same length and $\cdot$ indicates the operation of dot product between the two vectors.

Based on the cosine distance, the pairwise consistency can be maintained by the pairwise consistency loss, which is defined as

$$\mathcal{L}_{\text{pair}} = D\left(\varphi_i^I, \varphi_i^V\right) \tag{8}$$

where φ_i^I and φ_i^V denote the representations of the ith image and voice, respectively. Minimizing the pairwise consistency loss leads to a common space, where representations describing the same semantic concept are clustered together and the pairwise consistency is preserved.

3) Intramodality Consistency: The intramodality consistency measures the relationships between two representations from the same modality and belonging to the same semantic concept. In order to preserve the intramodality consistency, an intramodality consistency loss is introduced, which is defined as

$$\begin{aligned}\mathcal{L}_{\text{intra}} = {} & \hbar\left(1 - \ell_{ij}\left(\xi - D\left(\varphi_i^I, \varphi_j^I\right)\right)\right) \\ & + \hbar\left(1 - \ell_{ij}\left(\xi - D\left(\varphi_i^V, \varphi_j^V\right)\right)\right)\end{aligned} \tag{9}$$

where $\hbar(x) = \max(0, x)$ is the hinge function, ξ is the predefined margin threshold, and ℓ_{ij} denotes the label indicator. If φ_i^I and φ_j^I or φ_i^V and φ_j^V describe the same semantic concept, $\ell_{ij} = +1$; otherwise, $\ell_{ij} = -1$. It is to noted that $i \neq j$. Minimizing the intramodality consistency loss results in a common space, where representations describing the same semantic concept within the same modality are enforced to be close, whereas representations describing different semantic concepts within the same modality are enforced to be far.

4) Nonpaired Intermodality Consistency: The nonpaired intermodality consistency measures the relationships between two representations from different modalities but belonging to the same semantic concept. Since the pairwise loss only models the relationships of paired image–voice pairs with the same semantic concept in the applied RS image–voice data sets, the relationships between other nonpaired data cannot be explored effectively, and the compact representations for

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different semantic concepts are unable to be learned. To this end, the intermodality consistency loss is used to make up for the lack of the pairwise loss. To maintain the nonpaired intermodality consistency, the intermodality consistency loss is defined as

$$\mathcal{L}_{\text{inter}} = \hbar\left(1 - \ell_{ij}\left(\zeta - D\left(\varphi_i^V, \varphi_j^I\right)\right)\right) + \hbar\left(1 - \ell_{ij}\left(\zeta - D\left(\varphi_i^I, \varphi_j^V\right)\right)\right) \quad (10)$$

where ζ is the predefined margin threshold. If φ_i^V and φ_j^I or φ_i^I and φ_j^V describe the same semantic concept, $\ell_{ij} = +1$; otherwise, $\ell_{ij} = -1$. Minimizing the intermodality consistency loss leads to a common space, where representations describing the same semantic concept across the two modalities are enforced to be close, whereas representations describing different semantic concepts across the two modalities are enforced to be far.

D. Loss Function

1) Consistency Loss: The consistency loss is defined in the consistent representation space to model the similarity of the learned representations, which combines the pairwise, intramodality, and nonpaired intermodality consistency mentioned above. As a consequence, the consistent representation space possesses all the advantages that are contained in all three common spaces obtained by pairwise, intramodality, and intermodality consistency loss. Considering that the intramodality and intermodality losses are calculated from the same two pairs of samples, the significance of the intramodality loss and the intermodality consistency loss is set equal. Therefore, the consistency loss in the consistent representation space is defined as

$$\mathcal{L}_{\text{consi}} = \mathcal{L}_{\text{pair}} + \eta_1(\mathcal{L}_{\text{intra}} + \mathcal{L}_{\text{inter}}) \quad (11)$$

where η_1 is the tradeoff coefficient controlling the contribution of the last two terms.

2) Classification Loss: As for the representation on the top of each branch, a classification process is added to mine the intrinsic semantic information in each image and voice and further to model the discrimination and compactness of the learned representations. Specifically, a softmax layer is adopted for each image branch as

$$p_m^I = \text{softmax}\left(\mathbf{W}^I \varphi_m^I + \mathbf{b}^I\right) \quad (12)$$

where p_m^I is the probability of belonging to the corresponding semantic concept for the mth image, softmax represents the softmax activation function, and $\mathbf{W}^I$ and $\mathbf{b}^I$ are the parameters in the softmax layer. Similarly, a softmax layer is adopted for each voice branch as

$$p_m^V = \text{softmax}\left(\mathbf{W}^V \varphi_m^V + \mathbf{b}^V\right) \quad (13)$$

where p_m^V is the probability belonging to the corresponding semantic concept for the mth voice and $\mathbf{W}^V$ and $\mathbf{b}^V$ are the parameters in the softmax layer.

Then, the classification loss can be defined as

$$\mathcal{L}_{\text{class}} = -y_{it}\log\left(p_{it}^I + \epsilon\right) - y_{it}\log\left(p_{it}^V + \epsilon\right) \quad (14)$$

where y_{it} is the true semantic label of the ith sample, with t the tth class. p_{it}^I is the predicted probability distribution of the ith images. p_{it}^V is the predicted probability distribution of the ith voices, where t indexes the tth class. ϵ represents a regularization constant to avoid the NaN value in the loss. By minimizing the classification loss function, the semantic discrimination ability of the common representations in the consistent representation space can be greatly enhanced.

3) Joint Loss: With the above definition, a joint loss function is raised to simultaneously calculate the consistency loss of the consistent representation space and the classification loss of each branch. The joint loss function can be written as

$$\mathcal{L}_{\text{joint}} = \mathcal{L}_{\text{consi}} + \eta_2\mathcal{L}_{\text{class}} \quad (15)$$

where η_2 is the tradeoff coefficient controlling the contribution of the second term. By combining (11) and (15), the joint loss can be rewritten as

$$\mathcal{L}_{\text{joint}} = \mathcal{L}_{\text{pair}} + \eta_1(\mathcal{L}_{\text{intra}} + \mathcal{L}_{\text{inter}}) + \eta_2\mathcal{L}_{\text{class}}. \quad (16)$$

By minimizing the joint loss, the semantics-consistent representations are learned for the RS image–voice retrieval.

E. Optimizing Strategy

Based on the joint loss, the proposed SCRL method is optimized as follows. The parameter θ_I in the image encoding network is initialized by the VGG16 network pretrained on the ImageNet [43]. The parameters θ_V in the voice encoding network, Θ_I and Θ_V, in the semantics-consistent representation step are randomly initialized by the truncated_normal distribution [44]. In each training iteration, the optimizing process can be divided into five main steps. First, we sample two pairs of RS image–voice from the training collection of RS image–voice pairs. Second, the sampled RS image–voice pairs are input into the semantics encoding step to obtain image and voice features according to (3) and (4), respectively. Third, the extracted image and voice features are taken as an input of the semantics-consistent representation step for learning semantics-consistent representations via (5) and (6). Fourth, the joint loss is calculated according to (16). Finally, the parameters θ_V, Θ_I, and Θ_V are updated by minimizing the joint loss with the RMSprop optimization algorithm [45]. When a fixed iterative epoch, which is set as 50, is reached or the model is convergent, the optimizing process is terminated. Afterward, the parameters θ_I, θ_V, Θ_I, and Θ_V are leveraged to compute ultimate semantics-consistent representations for the RS image–voice retrieval. Note that the proposed SCRL method is trained end-to-end. The details about the optimization process of the proposed SCRL method are shown in Algorithm 1.

IV. Experiment and Results

In this section, experimental data sets, implementation details, evaluation metrics, and parameter analysis are introduced. In addition, the experimental results are compared with some state-of-the-art methods and the ablation analysis is presented to prove the effectiveness of the proposed SCRL method.

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Algorithm 1 The Proposed SCRL Method

Input:

Training collection $\mathcal{D} = \left\{\left(\mathbf{X}_m^I, \mathbf{X}_m^V, \mathbf{y}_m\right)\right\}_{m=1}^N$ of RS image-voice pairs;

Learning rate lr, trade-off coefficients η_1 and η_2, iterative epoch K.

Output:

The parameters θ_I and θ_V in the feature extraction step;

The parameters Θ_I and Θ_V in the representation learning step.

Initialization:

The parameter θ_I is initialized by the pre-trained VGG16 network;

The parameters θ_V, Θ_I and Θ_V are randomly initialized by truncated_normal distribution.

Repeat:

1: Sample the input RS image-voice pairs $\left(\mathbf{X}_i^I, \mathbf{X}_i^V, \mathbf{y}_i\right)$ and $\left(\mathbf{X}_j^I, \mathbf{X}_j^V, \mathbf{y}_j\right)$;

2: Calculate the image features $\mathbf{s}_i^I$ and $\mathbf{s}_j^I$ via Eq. 3, and voice features $\mathbf{s}_i^V$ and $\mathbf{s}_j^V$ according to Eq. 4;

3: Learn the semantics-consistent representations φ_i^I and φ_j^I according to Eq. 5, as well as φ_i^V and φ_j^V via Eq. 6;

4: Calculate the joint loss $\mathcal{L}_{joint}$ according to Eq. 16;

5: Update the parameters θ_V, Θ_I and Θ_V by utilizing RMSProp.

Until: A fixed iterative epoch K or convergence.

Return: θ_I, θ_V, Θ_I and Θ_V

A. Data Sets

In order to verify the proposed SCRL method, three challenging RS image–voice data sets are applied, including Sydney, UCM, and RSICD image–voice data sets [10]. A brief introduction about the applied RS image–voice data sets is given as follows.

1) Sydney Image–Voice Data Set [10]: The Sydney image–voice (Sydney IV) data set contains 613 RS images and 3065 voices of seven classes, where each image corresponds to five different voices. In this article, we follow the previous works [11]–[13] and randomly sample a voice from the five voices for each image to construct RS image–voice pairs. As for the data partitioning, 80% RS image–voice pairs are randomly selected for training, and the rest 20% are selected for testing.

2) UCM Image–Voice Data Set [10]: The UCM image–voice (UCM IV) data set includes 2100 RS images and 10500 voices, where each image corresponds to five different voices. Note that the data set can be divided into 21 classes, where each class includes 100 images and 500 voices. In this article, we follow the previous works [11]–[13] and randomly sample a voice from the five voices for each image to construct RS image–voice pairs. As for the data partitioning, 80% RS image–voice pairs are randomly selected as training set, and the rest 20% are selected as testing set.

3) RSICD Image–Voice Data Set [10]: The RSICD image–voice (RSICD IV) data set involves 10921 RS images and 54605 voices of 30 classes, where each image corresponds to five different voices. In this article, we follow the previous works [11]–[13] and randomly sample a voice from the five voices for each image to construct RS image–voice pairs. As for the data partitioning, 80% RS image–voice pairs are randomly selected for training, and the rest 20% are selected for testing.

TABLE I

DETAILED ARCHITECTURE OF THE VOICE NETWORK

Layer	Output-Size	Kernel	Stride
Conv1	24000×1×1	7×1×1	1×1×1
Conv2	12000×1×6	3×1×6	1×2×1
Conv3	6000×1×12	3×1×12	1×2×1
Pool1	3000×1×12	1×2×1	1×2×1
Conv4	1500×1×24	3×1×24	1×2×1
Conv5	750×1×48	3×1×48	1×2×1
Pool2	375×1×48	1×2×1	1×2×1

B. Implementation Details

In this work, the proposed SCRL method contains four branches, in which two branches are responsible for the image modality, and the others are responsible for the voice modality. As for branches, responsible for the same modality, the parameters are sharing. The specific architecture of the voice encoding network is reported in Table I. The dilation rate in the first convolutional layer is set as 3 and the others are set as 2. The input images are uniformly adjusted to 224 × 224. Before input into the voice network, the voices are resampled at 22 050 Hz and preprocessed as MFCC features by a window with size of 16 ms and shift of 5 ms. From the output of the semantics encoding step, the image features are with size of 7 × 7 × 512 (namely, $h_1 = w_1 = 7$ and $c_1 = 512$), and the voice features are with size of 375 × 1 × 48 (namely, $h_2 = 375$, $w_2 = 1$, and $c_2 = 48$). Before input into the SCRL step, the image features are adjusted as size of 512 by the operation of global average pooling [46], and the voice features are adjusted as size of 18 000 by the operation of flattening. In the semantics-consistent representation step, ReLU and Tanh are applied as activation functions of image branches and voice branches, respectively [47], [48]. The ultimate semantics-consistent representations are with size of 1024 (namely, $d_1 = d_2 = 1024$).

The proposed method is optimized utilizing an RMSProp optimizer [44], in which the weight decay is set as 0.0005 and the momentum is set to 0.9. The learning rate is set as 0.0004. The batch size is set to 32. The tradeoff coefficients η_1 and η_2 are set as 1 and 0.1, respectively, which are determined by a grid search strategy [13]. The experiment is conducted on the PC with a TITAN X (Pascal) GPU and 64-GB RAM.

C. Evaluation Metrics and Comparison Methods

In this work, two protocols are adopted, including using images as queried samples to retrieve the corresponding voices (I → V) and using voices as queried samples to retrieve the corresponding images (V → I). To evaluate the performance of the proposed SCRL method, two commonly used evaluation

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metrics are adopted, including mAP [49] and the precision of the top-k ranking result (**P@k**) [50]. The mAP score measures the mean value of average precision, which considers the precision and the returned ranking results at the same time. Precision is the ratio of the returned relevant samples to the queried samples. The **P@k** score measures the precision of the top-k retrieved samples. In this article, the **P@k** score is reported when k equals 1, 5, and 10, which is denoted as **P@1**, **P@5**, and **P@10**, respectively. In addition, the precision curve is shown when the number of retrieved samples changes to further evaluate the proposed SCRL method.

In order to assess the effectiveness of the proposed SCRL method, seven comparison methods are adopted, including SIFT + M [51], DBLP [52], CNN + SPEC [53], DVAN [10], CMIR-NET [7], DIVR [13], and DTBH [12] methods. The SIFT + M method [51] leverages the SIFT features of images and MFCC features of voices to perform the RS image–voice retrieval. The DBLP method [52] adopts an unsupervised manner to learn the coherence between audio and visual modalities. The CNN + SPEC method [53] seamlessly unifies the learning of different modalities in an unsupervised manner. The DVAN method [10] presents a novel image–voice learning framework to learn the cross-modal similarity. The CMIR-NET method [7] proposes to learn the discriminative shared feature space of the input data for RS image–voice retrieval. The DIVR method [13] leverages multiscale context information guiding the low-dimensional hash code for RS image–voice retrieval. The DTBH method [12] establishes a deep triplet-based hashing method, which integrates the hash code learning and the representation learning into a unified framework. These methods are implemented in this work. It is to note that the hash-based retrieval methods, including DTBH and DIVR methods, use a 64-bit hash code for comparison. The experimental results and the corresponding analysis are given as follows.

D. Parameter Analysis

The hyperparameter η_1 controls the contributions of the intramodality and nonpaired intermodality consistency, and η_2 controls the contribution of modeling the semantic classification. To study the performance impacts of them, we conduct parameter experiments about the two parameters with the I $\rightarrow$ V protocol on the Sydney IV data set. To this end, a grid search strategy is adopted to tune the two parameters following the previous work [54]. Specially, the range of η_1 and η_2 is set as $\{0, 0.2, 0.4, 0.8, 1.0, 1.2, 1.4, 1.6, 1.8, 2.0\}$ and $\{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10\}$, respectively. Note that $\eta_1 = 0$ represents the SCRL with $\mathcal{L}_{\text{pair}}$ and $\mathcal{L}_{\text{class}}$ only, and $\eta_2 = 0$ represents the SCRL with $\mathcal{L}_{\text{consi}}$ only. The parameter experiments are conducted by changing one hyperparameter (e.g., η_1) while fixing the other (e.g., η_2). The detailed implementation and analysis are as follows.

First, the value of η_2 is fixed at 1 empirically, and η_1 is changed from 0 to 2 with increment 0.2 per step. The corresponding mAP scores about the parameter η_1 are recorded and shown in Fig. 3. From the figure, it can be seen that both lower and higher values for η_1 result in lower mAP value, and the best mAP is obtained when η_1 equals 1. Notice that when $\eta_1 = 0$, the retrieval performance of the proposed method is significantly reduced because the neglect of the intramodality consistency and nonpaired intermodality consistency leads to the same semantic concept across the two modalities not being matched correctly.

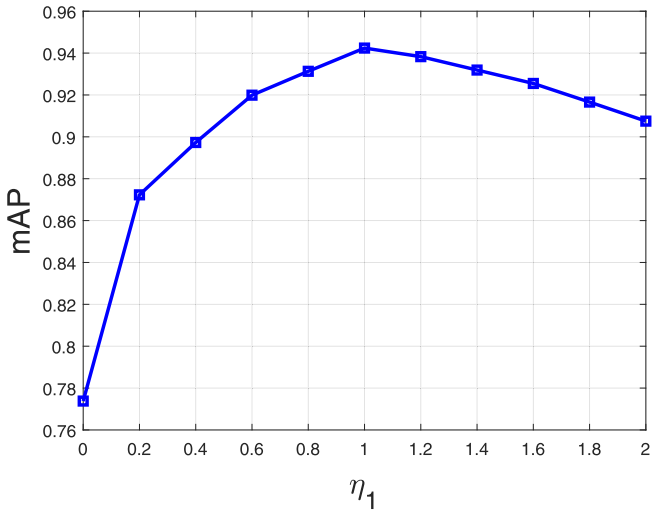


Fig. 3. Retrieval performance with different values for η_1 by the I $\rightarrow$ V protocol on the Sydney IV data set.

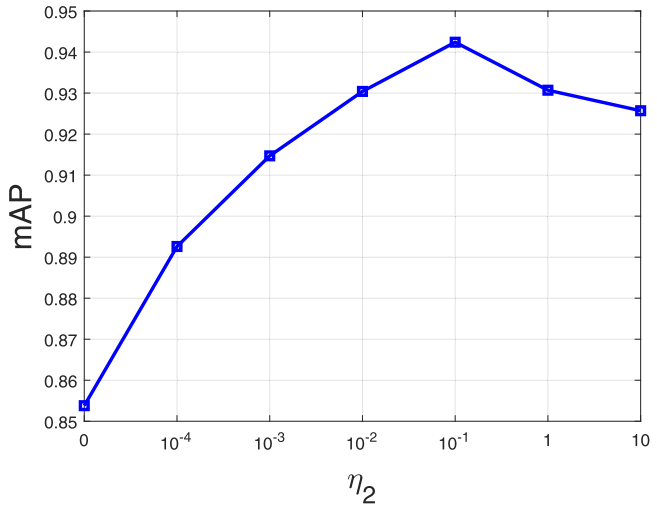


Fig. 4. Retrieval performance with different values for η_2 by the I $\rightarrow$ V protocol on the Sydney IV data set.

Second, the value of η_1 is fixed at 1, η_2 is changed from 0 to 10 with a ten times increment each step. Fig. 4 shows the performance for different η_2 values. From the figure, we can observe that the best performance is obtained when $\eta_2 = 0.1$. It is noted that when η_2 equals 0, the retrieval performance gets worse slightly, which indicates that the classification loss contributes to enhancing the semantic discrimination ability and compactness of the semantics-consistent representations.

As a result, the tradeoff coefficients η_1 and η_2 in the proposed SCRL method are set to 1 and 0.1, respectively.

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TABLE II

COMPARISON RESULTS BETWEEN THE PROPOSED SCRL METHOD AND OTHER METHODS ON THE SYDNEY IV DATA SET

Protocols	Methods	mAP	**P@1**	**P@5**	**P@10**
I→V	SIFT+M [51]	31.67	11.21	35.00	37.59
	DBLP [52]	44.38	56.51	52.65	49.68
	CNN+SPEC [53]	46.67	58.62	55.00	51.64
	DVAN [10]	71.77	75.86	73.62	72.93
	CMIR-NET [7]	78.44	84.68	82.54	81.04
	DIVR [13]	81.35	88.26	86.35	84.47
	DTBH [12]	92.45	97.41	95.63	93.78
	SCRL-Pair	75.80	80.11	75.07	71.90
	SCRL-Intra	79.79	85.43	81.20	74.78
	SCRL-Inter	80.19	86.24	82.50	76.57
	SCRL-Class	85.38	90.87	88.11	85.99
	SCRL-Dilation	92.08	93.68	93.03	91.65
	SCRL	**94.24**	**95.50**	**95.17**	**93.95**
V→I	SIFT+M [51]	26.50	34.48	24.48	23.28
	DBLP [52]	34.87	21.63	26.78	30.94
	CNN+SPEC [53]	35.72	17.24	27.76	31.21
	DVAN [10]	63.88	67.24	63.34	67.07
	CMIR-NET [7]	71.28	76.69	74.52	71.60
	DIVR [13]	75.97	80.44	78.05	76.27
	DTBH [12]	87.49	92.18	90.36	88.82
	SCRL-Pair	75.19	78.67	76.75	75.12
	SCRL-Intra	79.96	85.37	84.55	83.17
	SCRL-Inter	80.52	85.99	85.15	83.69
	SCRL-Class	84.87	88.62	87.64	86.75
	SCRL-Dilation	91.14	92.68	89.59	89.11
	SCRL	**93.53**	**94.31**	**91.38**	**90.00**

TABLE III

COMPARISON RESULTS BETWEEN THE PROPOSED SCRL METHOD AND OTHER METHIDS ON THE UCM IV DATA SET

Protocols	Methods	mAP	**P@1**	**P@5**	**P@10**
I→V	SIFT+M [51]	8.55	4.56	4.65	4.56
	DBLP [52]	25.48	24.18	23.87	23.24
	CNN+SPEC [53]	26.25	29.50	25.52	23.65
	DVAN [10]	36.79	32.37	33.29	33.74
	CMIR-NET [7]	45.82	52.92	49.74	43.38
	DIVR [13]	50.94	59.34	54.17	50.12
	DTBH [12]	64.24	73.10	69.69	65.63
	SCRL-Pair	35.25	35.24	35.48	35.50
	SCRL-Intra	52.64	59.90	57.10	54.19
	SCRL-Inter	53.62	61.95	56.33	52.17
	SCRL-Class	60.54	68.81	64.86	59.62
	SCRL-Dilation	65.46	72.57	70.43	67.86
	SCRL	**67.97**	**77.38**	**75.29**	**72.26**
V→I	SIFT+M [51]	6.66	3.58	4.41	4.68
	DBLP [52]	19.33	17.12	17.62	16.31
	CNN+SPEC [53]	21.79	19.42	19.86	19.23
	DVAN [10]	32.28	32.37	33.91	34.34
	CMIR-NET [7]	40.37	46.74	43.75	39.62
	DIVR [13]	45.34	52.23	48.76	44.98
	DTBH [12]	60.13	70.26	66.63	61.73
	SCRL-Pair	34.69	39.76	37.38	36.76
	SCRL-Intra	50.74	56.67	52.86	48.40
	SCRL-Inter	49.31	56.86	53.86	51.90
	SCRL-Class	59.82	66.90	59.05	55.76
	SCRL-Dilation	66.59	72.62	68.52	64.31
	SCRL	**66.83**	**84.05**	**77.14**	**74.07**

E. Ablation Analysis

In this section, five variations of the proposed SCRL method are conducted on three challenging RS image–voice data sets to examine: 1) the importance of the pairwise consistency loss; 2) the impact of the intramodality consistency loss; 3) the importance of the intermodality consistency loss; 4) the importance of the classification loss; and 5) the effect of the dilated convolutional kernel. The details about the implementation of different variations are elaborated as follows. First, the variation of the proposed SCRL method without the pairwise consistency loss (SCRL-Pair) is implemented to verify the effect of the pairwise consistency loss. Second, the intramodality consistency loss is discarded evolving into a new variation (SCRL-Intra). Third, we omit the intermodality consistency loss obtaining another new variation (SCRL-Inter) to determine the importance of the nonpaired intermodality relationship. Fourth, the classification loss is abandoned evolving into the variation SCRL-Class. Finally, the dilation rate in the voice network is set as 1 generating the new variation SCRL-Dilation. The results of these variations are reported in Tables II–IV. According to the results, we give the analysis as the following aspects.

1) Pairwise Consistency: The pairwise consistency loss aims to narrow the distance between the representations from the RS image–voice pair. From the results in Tables II–IV, we can intuitively find that the retrieval performance is greatly improved when the pairwise consistency is considered. Concretely, compared with the method without the pairwise consistency loss (SCRL-Pair), the proposed SCRL method can improve the mAP value from 75.80% to 94.24%, from 35.25% to 67.97%, and from 15.80% to 28.84% for the I → V protocol on the Sydney, UCM, and RSICD IV data sets, respectively. For the V → I protocol, the mAP value is also improved greatly by the proposed SCRL method compared with the SCRL-Pair method. The improvements demonstrate that the pairwise consistency loss is effective for learning semantics-consistent representations.

2) Intramodality Consistency: The intramodality consistency loss aims to narrow the distance between two representations in the same modality to keep the semantic consistency of the same semantic concept. From the results in Tables II–IV, we can intuitively find that the retrieval performance is greatly improved when the intramodality consistency is considered. The specific comparison situation corresponds to the results of SCRL-Intra method and the proposed SCRL method in Tables II–IV. The improvement is because the intramodality consistency loss can constrain the model to shorten the distance between two representations from the same semantic concept in each modality.

3) Nonpaired Intermodality Consistency: The nonpaired intermodality consistency aims to learn the relationship between two representations in different modalities to keep the semantic consistency across the two modalities. The specific comparison situation corresponds to the results of SCRL-Inter method and the proposed SCRL method in Tables II–IV. From the results, we can notice that the retrieval performance is

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TABLE IV
COMPARISON RESULTS BETWEEN THE PROPOSED SCRL METHOD AND OTHER METHODS ON THE RSICD IV DATA SET (%)

Protocols	Methods	mAP	P@1	P@5	P@10
I→V	SIFT+M [51]	5.04	6.22	5.34	4.50
	DBLP [52]	12.70	15.32	15.21	14.22
	CNN+SPEC [53]	13.24	16.82	16.62	15.69
	DVAN [10]	16.29	22.49	22.56	21.70
	CMIR-NET [7]	17.78	24.11	23.52	22.54
	DIVR [13]	19.62	25.43	24.84	24.20
	DTBH [12]	23.46	27.58	26.84	25.49
	SCRL-Pair	15.80	17.09	16.84	16.05
	SCRL-Intra	20.06	21.78	21.34	20.75
	SCRL-Inter	20.02	22.46	22.44	21.73
	SCRL-Class	25.92	29.54	28.35	27.00
	SCRL-Dilation	27.54	30.88	29.96	29.81
	SCRL	**28.84**	**31.10**	**29.81**	**28.34**
V→I	SIFT+M [51]	4.85	3.66	3.60	3.541
	DBLP [52]	8.14	6.21	6.08	6.76
	CNN+SPEC [53]	9.96	7.13	7.00	7.44
	DVAN [10]	15.71	16.18	15.10	14.76
	CMIR-NET [7]	17.25	17.94	16.58	15.36
	DIVR [13]	18.58	19.76	18.31	17.59
	DTBH [12]	22.72	23.30	22.48	21.17
	SCRL-Pair	15.35	18.56	17.31	16.35
	SCRL-Intra	24.52	30.11	29.08	28.08
	SCRL-Inter	25.21	30.60	30.72	29.64
	SCRL-Class	28.33	31.73	31.57	30.83
	SCRL-Dilation	29.55	32.59	31.25	30.08
	SCRL	**31.26**	**33.76**	**33.01**	**32.01**

significantly improved when the intramodality consistency is considered. This is because the intramodality consistency loss contributes to narrowing the distance between two representations from different modalities when they describe the same semantic concept.

4) Semantic Discrimination: The classification loss aims to enhance the semantic discrimination ability of representations so as to retrieve the relevant samples more easily. The specific comparison situation corresponds to the results of SCRL-Class method and the proposed SCRL method in Tables II–IV. From the comparison results, we can see that the scores of all evaluation metrics are lifted when the classification loss is added. This proves that the classification loss contributes to enhancing the semantic discrimination ability of representations.

5) Long-Range Correlation: The dilation convolution is adopted to capture the long-range correlation of each voice sample. The concrete comparison situation corresponds to the results of SCRL-Dilation method and the proposed SCRL method in Tables II–IV. The results show that the retrieval performance is slightly improved, which indicates that the dilation convolution can indeed capture the long-range correlation relationship within each voice sample by increasing the receptive field.

F. Results and Analysis

The experimental results and the corresponding analysis on three challenging RS image–voice data sets are given as follows.

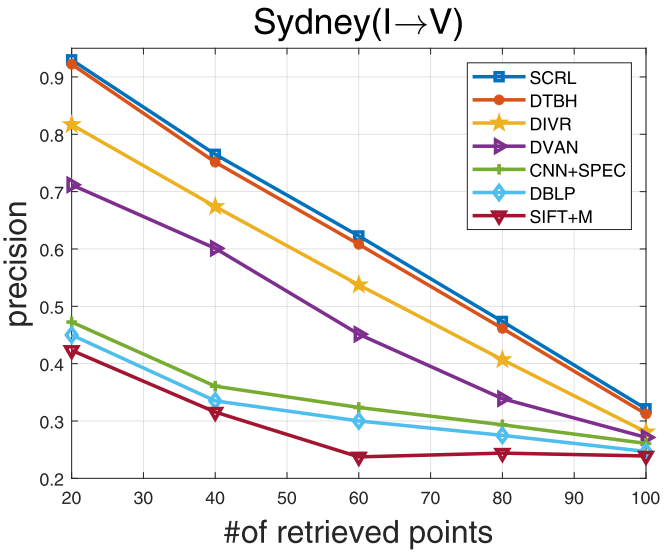


Fig. 5. Precision curves with different retrieved samples by the I → V protocol on the Sydney IV data set.

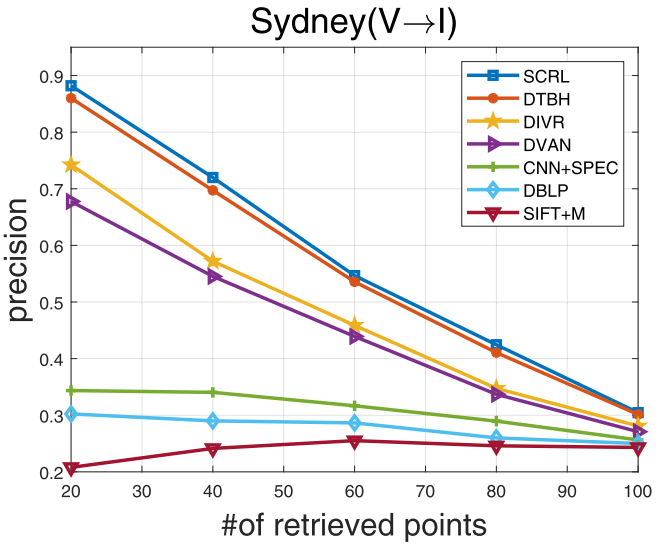


Fig. 6. Precision curves with different retrieved samples by the V → I protocol on the Sydney IV data set.

1) Results on the Sydney IV Data Set: Table II shows the comparison results between the proposed SCRL method and other compared methods on the Sydney IV data set. Figs. 5 and 6 show the precision curves with different retrieved samples by the I → V and V → I protocol, respectively. By observing the results in Table II and Figs. 5 and 6, we can find that: 1) the proposed SCRL method achieves the highest value in terms of most evaluation metrics; 2) Figs. 5 and 6 show that the proposed SCRL method outperforms other comparison methods at all returned neighbors; and 3) as for the I → V protocol, the proposed SCRL method improves the mAP value from SIFT + M (31.67%), DBLP (44.38%), CNN + SPEC (46.67%), DVAN (71.77%), CMIR-NET (78.44%), DIVR (81.35%), and DTBH (92.45%) to 94.24%. Meanwhile, for the V → I protocol, the proposed SCRL

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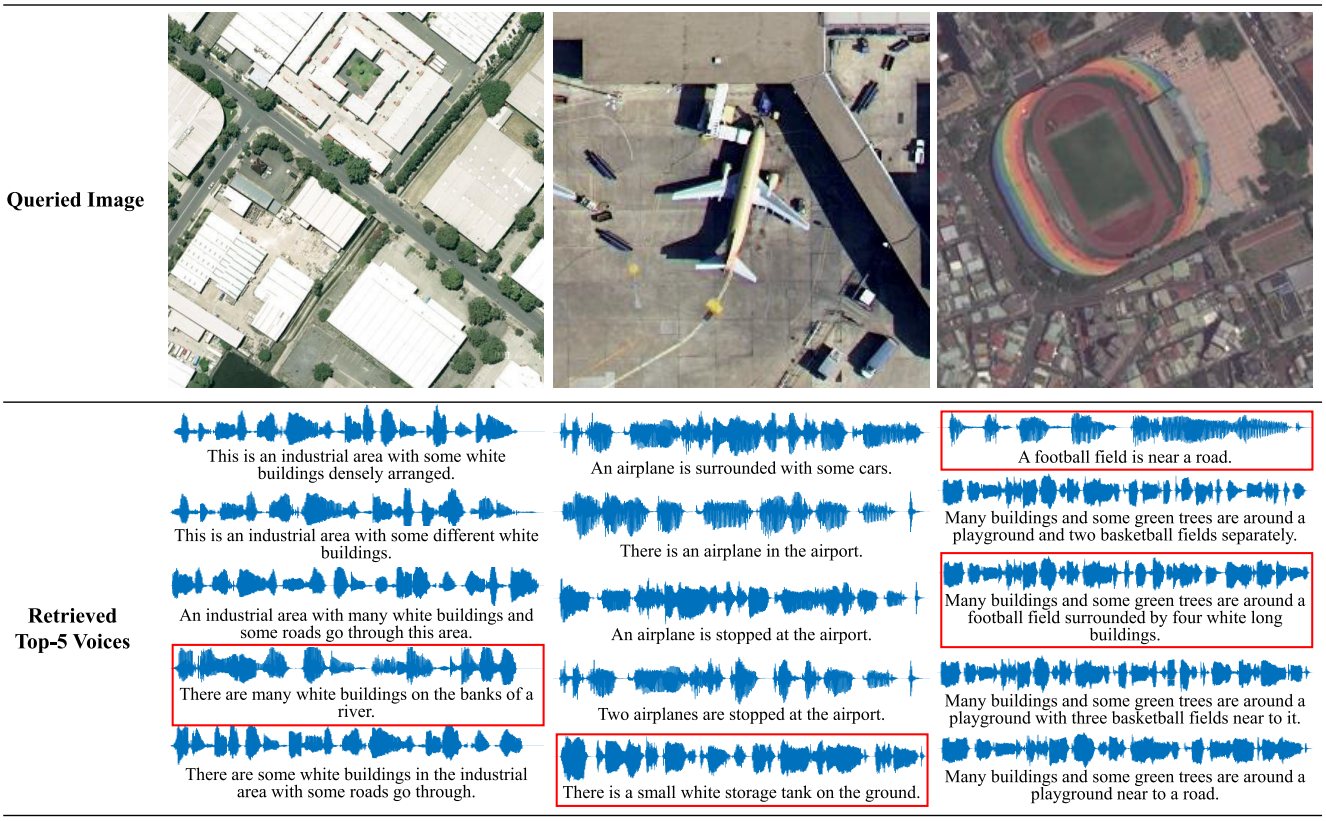


Fig. 7. Some examples of the I → V retrieval results by the proposed SCRL method on the Sydney, UCM, and RSICD IV data sets. The result examples on the three data sets correspond to the first column, second column, and third column, respectively. The wrong retrieval results are marked with red boxes.

method improves the mAP value from SIFT + M (26.50%), DBLP (34.87%), CNN + SPEC (35.72%), DVAN (63.88%), CMIR-NET (71.28%), DIVR (75.97%), and DTBH (87.49%) to 93.53%. The improvements demonstrate that the modeling for the pairwise, intramodality, and nonpaired intermodality relationships contributes to learning semantics-consistency representations across the two modalities, thereby facilitating cross-modal retrieval. In addition, the first column of Fig. 7 shows an example of the top five retrieved results by the proposed SCRL method with the I → V protocol. The first row of Fig. 8 shows an example of the top five retrieved results by the proposed SCRL method with the V → I protocol. As shown in the two figures, the proposed SCRL method can retrieve the relevant samples effectively, which further proves the effectiveness for modeling the semantics-consistent relationships comprehensively.

2) Results on the UCM IV Data Set: Table III shows the comparison results between the proposed SCRL method and other compared methods on the UCM IV data set. Figs. 9 and 10 show the precision curves with different retrieved samples by the I → V and V → I protocol, respectively. Similar experimental results can be seen on the Sydney IV data set. For instance, for the I → V protocol, the proposed SCRL method improves the mAP value from SIFT + M (31.67%), DBLP (44.38%), CNN + SPEC (46.67%), DVAN (71.77%), CMIR-NET (78.44%), DIVR (81.35%), and DTBH (92.45%) to 67.97%. Meanwhile, for the V → I protocol, the proposed SCRL method improves the mAP value from SIFT + M (26.50%), DBLP (34.87%), CNN + SPEC (35.72%), DVAN (63.88%), CMIR-NET (71.28%), DIVR (75.97%), and DTBH (87.49%) to 66.83%. This is

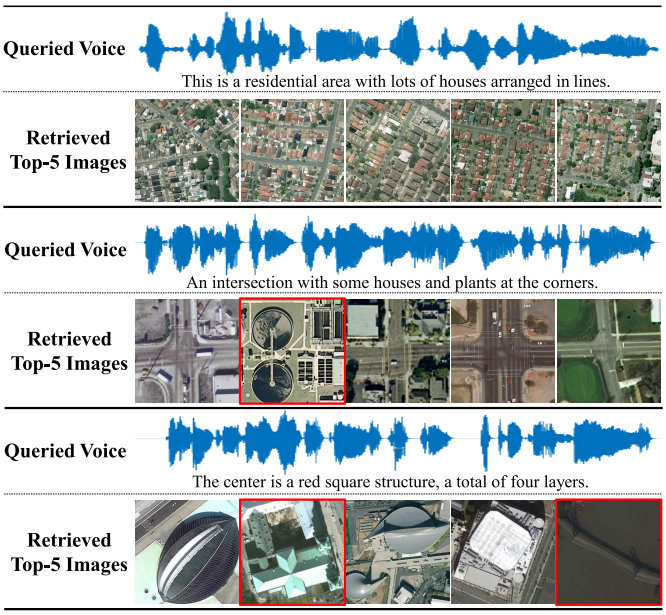


Fig. 8. Some examples of the V → I retrieval results by the proposed SCRL method on the Sydney, UCM, and RSICD IV data sets. The result examples on the three data sets correspond to the first row, second row, and third row, respectively. The wrong retrieval results are marked with red boxes.

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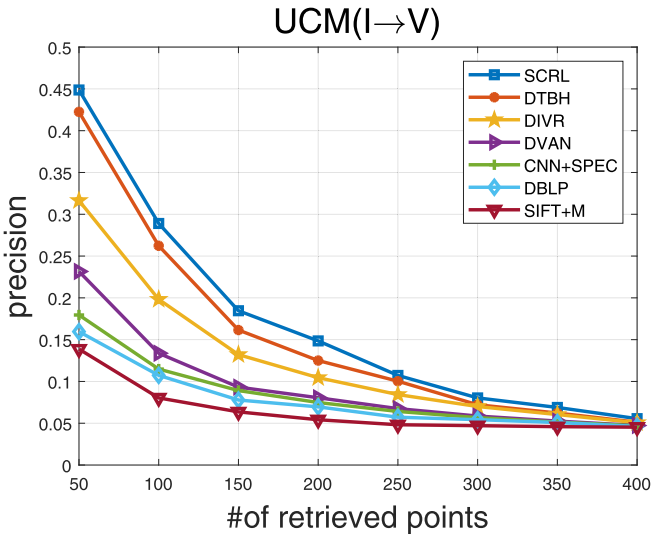


Fig. 9. Precision curves with different retrieved samples by the I → V protocol on the UCM IV data set.

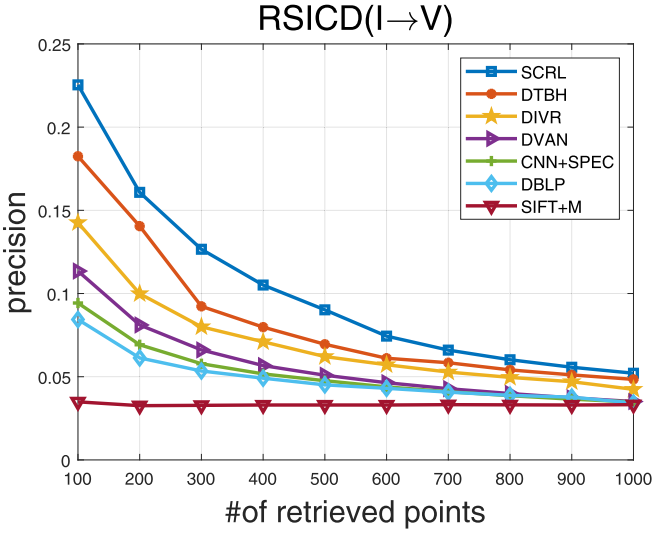


Fig. 11. Precision curves with different retrieved samples by the I → V protocol on the RSICD IV data set.

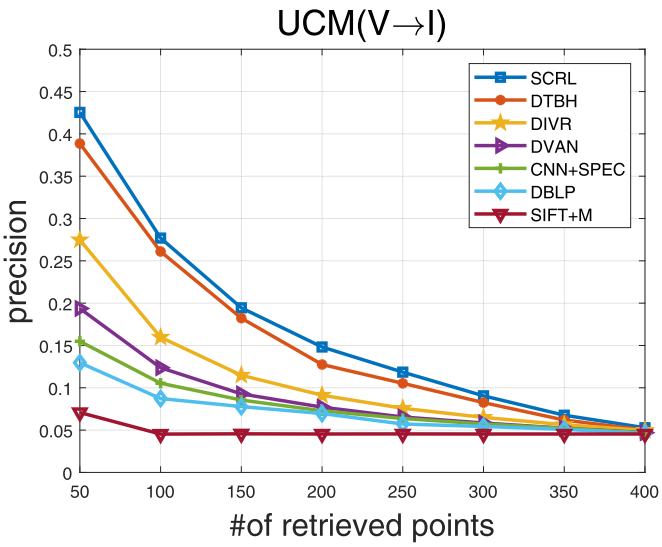


Fig. 10. Precision curves with different retrieved samples by the V → I protocol on the UCM IV data set.

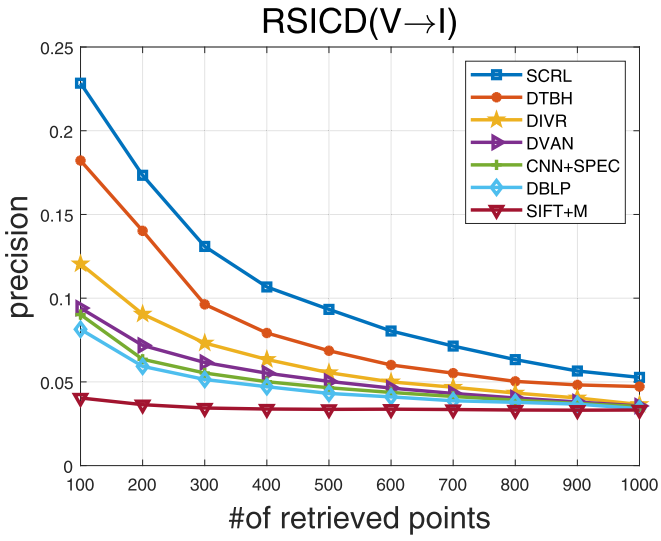


Fig. 12. Precision curves with different retrieved samples by the V → I protocol on the RSICD IV data set.

because the exploration for the semantics-consistent representation space can effectively promote the model to narrow the heterogeneous semantic gap across the two modalities. In addition, the second column of Fig. 7 shows an example of the top five retrieved results by the proposed SCRL method with the I → V protocol. The second row of Fig. 8 shows an example of the top five retrieved results by the proposed SCRL method with the V → I protocol. The results show that the proposed SCRL method can effectively retrieve relevant samples, which further proves the effectiveness of considering the pairwise, intramodality, and nonpaired intermodality relationships comprehensively.

3) Results on the RSICD IV Data Set: Table IV shows the comparison results between the proposed SCRL method and other compared methods on the most challenging RSICD IV data set. Figs. 11 and 12 show the precision curves with different retrieved samples by the I → V and V → I protocol, respectively. Although the RSICD IV data set is more challenging [13], the proposed SCRL method can improve the retrieval performance to a great extent. For example, the SCRL method lifts the mAP value from 23.46% to 28.84% for the I → V protocol compared with the state-of-the-art DTBH method. Meanwhile, the SCRL method improves the mAP value from 23.46% (DTBH) to 28.84% for the V → I protocol. This demonstrates the high efficiency of the proposed SCRL method for learning semantics-consistent representations across the two modalities because the semantics-consistent relationships, including pairwise, intramodality, and nonpaired intermodality relationships, are modeled comprehensively. In addition, some examples of the top five retrieved results by the proposed SCRL method are shown in Figs. 7 (third column) and 8 (third row). It can be seen that the

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proposed SCRL method can retrieve diverse but relevant samples effectively, even on very challenging image–voice data set, which further verifies the effectiveness of simultaneously modeling various semantics-consistent relationships.

V. Conclusion

In this article, an SCRL method is proposed for the task of RS image–voice retrieval. The main novelty is that the proposed method takes the pairwise, intramodality, and nonpaired intermodality relationships into account simultaneously, thereby improving the semantic consistency of the learned representations for the RS image–voice retrieval. Specifically, a consistent representation space is explored with a joint loss function comprehensively considering the three kinds of relationships for the extracted representations. By exploring the consistent representation space, the heterogeneous semantic gap can be mitigated to a great extent and semantics-consistent representations can be learned for the RS image–voice retrieval. Experimental results on the Sydney, UCM, and RSICD image–voice data sets demonstrate that the proposed joint loss is effective to mine the pairwise, intramodality, and nonpaired intermodality relationships for better RS image–voice retrieval. In addition, the proposed SCRL method achieves better performance compared with other state-of-the-art methods, manifesting its superiority. In the future, we will explore how to extend the proposed SCRL method to address the large-scale image–audio retrieval cases such as embedding the hashing code learning to our work.

References

[1] Y. Hua, L. Mou, and X. X. Zhu, "Relation network for multilabel aerial image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4558–4572, Jul. 2020.

[2] Y. Li, Y. Zhang, X. Huang, and J. Ma, "Learning source-invariant deep hashing convolutional neural networks for cross-source remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6521–6536, Nov. 2018.

[3] Y. Zhang, Y. Yuan, Y. Feng, and X. Lu, "Hierarchical and robust convolutional neural network for very high-resolution remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5535–5548, Aug. 2019.

[4] Y. Li, Y. Zhang, X. Huang, H. Zhu, and J. Ma, "Large-scale remote sensing image retrieval by deep hashing neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 950–965, Feb. 2018.

[5] B. Demir and L. Bruzzone, "Hashing-based scalable remote sensing image search and retrieval in large archives," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 2, pp. 892–904, Feb. 2016.

[6] Y. Chen, X. Lu, and X. Li, "Supervised deep hashing with a joint deep network," *Pattern Recognit.*, vol. 105, Sep. 2020, Art. no. 107368.

[7] U. Chaudhuri, B. Banerjee, A. Bhattacharya, and M. Datcu, "CMIR-NET: A deep learning based model for cross-modal retrieval in remote sensing," *Pattern Recognit. Lett.*, vol. 131, pp. 456–462, Mar. 2020.

[8] Z. Shi and Z. Zou, "Can a machine generate humanlike language descriptions for a remote sensing image?" *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3623–3634, Jun. 2017.

[9] Z. Shao, W. Zhou, X. Deng, M. Zhang, and Q. Cheng, "Multilabel remote sensing image retrieval based on fully convolutional network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 318–328, 2020.

[10] G. Mao, Y. Yuan, and L. Xiaoqiang, "Deep cross-modal retrieval for remote sensing image and audio," in *Proc. 10th IAPR Workshop Pattern Recognit. Remote Sens. (PRRS)*, Aug. 2018, pp. 1–7.

[11] M. Guo, C. Zhou, and J. Liu, "Jointly learning of visual and auditory: A new approach for RS image and audio cross-modal retrieval," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 11, pp. 4644–4654, Nov. 2019.

[12] Y. Chen and X. Lu, "A deep hashing technique for remote sensing image-sound retrieval," *Remote Sens.*, vol. 12, no. 1, p. 84, Dec. 2019.

[13] Y. Chen, X. Lu, and S. Wang, "Deep cross-modal image-voice retrieval in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7049–7061, Oct. 2020.

[14] M. Lin, R. Ji, S. Chen, X. Sun, and C.-W. Lin, "Similarity-preserving linkage hashing for online image retrieval," *IEEE Trans. Image Process.*, vol. 29, pp. 5289–5300, 2020.

[15] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*. [Online]. Available: https://arxiv.org/abs/1409.1556

[16] R. Hidayat, A. Bejo, S. Sumaryono, and A. Winursito, "Denoising speech for MFCC feature extraction using wavelet transformation in speech recognition system," in *Proc. 10th Int. Conf. Inf. Technol. Electr. Eng. (ICITEE)*, Jul. 2018, pp. 280–284.

[17] B. Luo, J.-F. Aujol, Y. Gousseau, and S. Ladjal, "Indexing of satellite images with different resolutions by wavelet features," *IEEE Trans. Image Process.*, vol. 17, no. 8, pp. 1465–1472, Aug. 2008.

[18] Y. Yang and S. Newsam, "Geographic image retrieval using local invariant features," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 2, pp. 818–832, Feb. 2013.

[19] E. Aptoula, "Remote sensing image retrieval with global morphological texture descriptors," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 3023–3034, May 2014.

[20] H. Ning, X. Zheng, Y. Yuan, and X. Lu, "Audio description from image by modal translation network," *Neurocomputing*, vol. 423, pp. 124–134, Jan. 2021.

[21] B. Zhao, L. Hua, X. Li, X. Lu, and Z. Wang, "Weather recognition via classification labels and weather-cue maps," *Pattern Recognit.*, vol. 95, pp. 272–284, Nov. 2019.

[22] Y. Yuan, J. Fang, X. Lu, and Y. Feng, "Remote sensing image scene classification using rearranged local features," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1779–1792, Mar. 2019.

[23] B. Zhao, X. Li, and X. Lu, "CAM-RNN: Co-attention model based RNN for video captioning," *IEEE Trans. Image Process.*, vol. 28, no. 11, pp. 5552–5565, Nov. 2019.

[24] Y. Yuan, J. Fang, X. Lu, and Y. Feng, "Spatial structure preserving feature pyramid network for semantic image segmentation," *ACM Trans. Multimedia Comput., Commun., Appl.*, vol. 15, no. 3, pp. 73:1–73:19, 2019.

[25] X. Tang, L. Jiao, W. J. Emery, F. Liu, and D. Zhang, "Two-stage reranking for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5798–5817, Oct. 2017.

[26] F. Ye, H. Xiao, X. Zhao, M. Dong, W. Luo, and W. Min, "Remote sensing image retrieval using convolutional neural network features and weighted distance," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 10, pp. 1535–1539, Oct. 2018.

[27] B. Chaudhuri, B. Demir, S. Chaudhuri, and L. Bruzzone, "Multilabel remote sensing image retrieval using a semisupervised graph-theoretic method," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 1144–1158, Feb. 2018.

[28] J. Kang, R. Fernandez-Beltran, D. Hong, J. Chanussot, and A. Plaza, "Graph relation network: Modeling relations between scenes for multilabel remote-sensing image classification and retrieval," *IEEE Trans. Geosci. Remote Sens.*, pp. 1–15, 2020.

[29] Y. Liu, L. Ding, C. Chen, and Y. Liu, "Similarity-based unsupervised deep transfer learning for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7872–7889, Nov. 2020.

[30] F. Ye, W. Luo, M. Dong, H. He, and W. Min, "SAR image retrieval based on unsupervised domain adaptation and clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 9, pp. 1482–1486, Sep. 2019.

[31] J. Song, T. He, L. Gao, X. Xu, A. Hanjalic, and H. T. Shen, "Unified binary generative adversarial network for image retrieval and compression," *Int. J. Comput. Vis.*, vol. 128, pp. 2243–2264, 2020.

[32] W. Xiong, Z. Xiong, Y. Cui, and Y. Lv, "A discriminative distillation network for cross-source remote sensing image retrieval," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1234–1247, 2020.

[33] W. Xiong, Z. Xiong, Y. Zhang, Y. Cui, and X. Gu, "A deep cross-modality hashing network for SAR and optical remote sensing images retrieval," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5284–5296, 2020.

[34] J. Song, Y. Yang, Y. Yang, Z. Huang, and H. T. Shen, "Inter-media hashing for large-scale retrieval from heterogeneous data sources," in *Proc. Int. Conf. Manage. Data SIGMOD*, 2013, pp. 785–796.

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[35] H. Noh, A. Araujo, J. Sim, T. Weyand, and B. Han, "Large-scale image retrieval with attentive deep local features," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 3456–3465.
[36] X. Lu, Y. Chen, and X. Li, "Hierarchical recurrent neural hashing for image retrieval with hierarchical convolutional features," *IEEE Trans. Image Process.*, vol. 27, no. 1, pp. 106–120, Jan. 2018.
[37] Y. Cui, Y. Song, C. Sun, A. Howard, and S. Belongie, "Large scale fine-grained categorization and domain-specific transfer learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4109–4118.
[38] Z. Chen, T. Zhang, and C. Ouyang, "End-to-end airplane detection using transfer learning in remote sensing images," *Remote Sens.*, vol. 10, no. 1, p. 139, Jan. 2018.
[39] L. A. Gatys, A. S. Ecker, and M. Bethge, "Image style transfer using convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2414–2423.
[40] A. Chowdhury and A. Ross, "Fusing MFCC and LPC features using 1D triplet CNN for speaker recognition in severely degraded audio signals," *IEEE Trans. Inf. Forensics Security*, vol. 15, pp. 1616–1629, 2020.
[41] Y. Chen and X. Lu, "Deep category-level and regularized hashing with global semantic similarity learning," *IEEE Trans. Cybern.*, early access, Feb. 25, 2020, doi: 10.1109/TCYB.2020.2964993.
[42] W. Xiong, Y. Lv, X. Zhang, and Y. Cui, "Learning to translate for cross-source remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4860–4874, Jul. 2020.
[43] Y. Chen and X. Lu, "Deep discrete hashing with pairwise correlation learning," *Neurocomputing*, vol. 385, pp. 111–121, Apr. 2020.
[44] Y. Yuan, H. Ning, and X. Lu, "Bio-inspired representation learning for visual attention prediction," *IEEE Trans. Cybern.*, early access, Sep. 2, 2019, doi: 10.1109/TCYB.2019.2931735.
[45] A. Yuan, X. Li, and X. Lu, "3G structure for image caption generation," *Neurocomputing*, vol. 330, pp. 17–28, Feb. 2019.
[46] T. Zhi, L.-Y. Duan, Y. Wang, and T. Huang, "Two-stage pooling of deep convolutional features for image retrieval," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2016, pp. 2465–2469.
[47] D. Hu, F. Nie, and X. Li, "Deep binary reconstruction for cross-modal hashing," *IEEE Trans. Multimedia*, vol. 21, no. 4, pp. 973–985, Apr. 2019.
[48] Z. Shao, Y. Pan, C. Diao, and J. Cai, "Cloud detection in remote sensing images based on multiscale features-convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 4062–4076, Jun. 2019.
[49] N. Khurshid, M. Tharani, M. Taj, and F. Z. Qureshi, "A residual-dyad encoder discriminator network for remote sensing image matching," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 2001–2014, Mar. 2020.
[50] P. Li and P. Ren, "Partial randomness hashing for large-scale remote sensing image retrieval," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 3, pp. 464–468, Mar. 2017.
[51] X. Zhao, Z. Li, and J. Yi, "SIFT feature-based second-order image hash retrieval approach," *J. Softw.*, vol. 13, no. 1, pp. 103–116, Feb. 2018.
[52] D. F. Harwath, A. Torralba, and J. R. Glass, "Unsupervised learning of spoken language with visual context," in *Proc. Adv. Neural Inf. Process. Syst.*, 2016, pp. 1858–1866.
[53] R. Arandjelovic and A. Zisserman, "Look, listen and learn," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 609–617.
[54] B. Wang, Y. Yang, X. Xu, A. Hanjalic, and H. T. Shen, "Adversarial cross-modal retrieval," in *Proc. 25th ACM Int. Conf. Multimedia*, Oct. 2017, pp. 154–162.



Hailong Ning is pursuing the Ph.D. degree in signal and information processing with the Shaanxi Key Laboratory of Ocean Optics, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, Shaanxi, China, and the University of Chinese Academy of Sciences, Beijing, China.

His main research interests include pattern recognition, machine learning, computer vision, and multimodal learning.



Bin Zhao is an Assistant Professor with the School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an, China.

His research interests include video summarization, video captioning, pattern recognition, and machine learning.

Yuan Yuan (Senior Member, IEEE) received the Ph.D. degree from the University of Bath, Bath, U.K., in 2005.

She is with the School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an, China. Her research interests include visual information processing and image/video content analysis.

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